



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Hotel Reservation Management System

The domain of the Project:

Salesforce Administration and Development

Team Mentors (and their designation):

Sri ram yarahbothula

Sales force developer

Submitted by:

Nithin Goli

Period of the project

August 2026-September 2026



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Declaration

The project titled “**Hotel Reservation Management System**” has been mentored by sri ram, organised by SURE Trust, from April 2023 to August 2023, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Submitted by

Mr.Nithin

Mentor's Name: Sri ram yarrabothula

Designation—Salesforce Developer

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

The **Hotel Reservation Management System** is a Salesforce-based application developed to simplify and centralize hotel management and reservation activities. The main objectives of the project are to manage hotels, rooms, guests, reservations, staff, room services, and payments efficiently while maintaining data accuracy and security.

The project was developed using Salesforce features such as **Custom Objects, Relationships, Validation Rules, Record Types, Salesforce Flow, Apex, Lightning Web Components, Reports, and Dashboards**. Automation was implemented for reservation confirmation, payment-based status updates, reservation completion, and check-in reminders. An Apex trigger was also used to prevent duplicate or overlapping room bookings.

The implemented system successfully provides a centralized platform for managing hotel operations and monitoring important information such as **room availability, occupancy, reservations, revenue, and staff performance** through reports and dashboards.

The project can be further enhanced with **online payment integration, customer self-service, mobile application support, advanced analytics, and AI-based hotel or room recommendations**.



Introduction

Background and Context

The **Hotel Reservation Management System** is a Salesforce-based application developed to manage hotel operations in a centralized and organized manner. Hotels need to manage different activities such as room availability, guest information, reservations, staff, room services, and payments. Managing these activities manually can be time-consuming and may lead to errors. The project uses Salesforce to bring these operations together in a single system.

Problem Statement / Goals

The main problem addressed by this project is the difficulty of managing hotel reservation activities efficiently while maintaining accurate and up-to-date information. The goals of the project are to:

- Manage hotel and room information.
- Maintain guest and reservation details.
- Track room availability and reservation status.
- Manage staff and room services.
- Track reservation payments.
- Prevent duplicate or overlapping room bookings.
- Automate important hotel processes.
- Provide reports and dashboards for better monitoring and decision-making.

Scope and Limitations

The scope of the project includes **Hotel, Room, Guest, Reservation, Staff, Room Service, and Payment management**. It also covers Salesforce automation using Flows, Apex-based business logic, Lightning Web Components, reports, dashboards, validation, and role-based security.

The current system is developed as a **Salesforce application** and does not include direct integration with external hotel booking platforms, online payment gateways, SMS services, or a dedicated mobile application. These features can be considered for future development.

Innovation Component

The innovation of the project is the combination of **Salesforce automation, Apex, and Lightning Web Components** to create an integrated hotel management solution. The system automatically handles important activities such as reservation confirmation, payment-based status updates, reservation completion, and check-in reminders. The **duplicate room booking prevention mechanism** improves reservation accuracy, while dashboards and reports provide useful insights into hotel operations and performance.



Project Objectives

Objectives and Goals

The main objectives of the **Hotel Reservation Management System** are:

- To develop a centralized system for managing hotel information.
- To manage room details, room types, prices, and availability.
- To maintain guest and reservation information efficiently.
- To manage check-in and check-out dates and reservation status.
- To manage hotel staff and room service activities.
- To track payments related to hotel reservations.
- To prevent duplicate or overlapping room bookings.
- To automate reservation and payment-related processes using Salesforce Flow.
- To improve data accuracy through validation rules and Apex.
- To provide interactive functionality using Lightning Web Components.
- To generate reports and dashboards for monitoring hotel operations.
- To implement role-based security for controlled access to hotel data.

Expected Outcomes and Deliverables

The expected outcomes and deliverables of the project are:

- A functional **Salesforce Hotel Reservation Management System**.
- Properly configured standard and custom objects with relationships.
- Validation rules for maintaining data accuracy.
- Record Types for different reservation and room categories.
- Automated Salesforce Flows for hotel and reservation processes.
- Apex Trigger for preventing duplicate room bookings.
- Batch Apex for occupancy-related processing.
- Lightning Web Components for dashboard, reservation, and staff management.
- Reports for rooms, occupancy, revenue, staff performance, and reservations.
- A centralized **Hotel Reservation Dashboard** for management insights.
- Improved efficiency, accuracy, and automation of hotel reservation operations



Methodology and Results

Methods / Technology Used

The project was developed using a structured Salesforce development approach. The following methods and technologies were used:

- Salesforce Data Modeling for designing objects and relationships.
- Custom Objects for managing Rooms, Reservations, Staff, Room Services, and Payments.
- Standard Objects such as Account and Contact for Hotels and Guests.
- Validation Rules for maintaining data accuracy.
- Record Types for managing different room and reservation categories.
- Salesforce Flow for automating hotel and reservation processes.
- Apex Trigger for preventing duplicate or overlapping room bookings.
- Batch Apex for occupancy-related processing.
- Lightning Web Components (LWC) for interactive application interfaces.
- Salesforce Reports and Dashboards for analyzing hotel operations.

Tools / Software Used

- **Salesforce Lightning Platform**
- **Salesforce Setup**
- **Salesforce Flow Builder**
- **Salesforce Developer Console**
- **Visual Studio Code**
- **Salesforce Extension Pack**
- **Salesforce CLI**
- **Lightning Web Components**
- **Apex**
- **GitHub** for project source-code management

Data Collection Approach

The project primarily uses data entered and managed within the Salesforce application. Test data such as **hotel details, room information, guest details, reservations, staff details, and payment records** was created to test the functionality of the system.



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The created data was used to verify reservation processing, room availability, payment-related updates, duplicate booking prevention, reports, dashboards, and automation.

Project Architecture

The system follows a Salesforce-based layered architecture:

User Interface Layer → Business Logic & Automation Layer → Data Layer → Reporting & Analytics

1. User Interface Layer

- Salesforce Lightning interface
- Lightning Web Components
- Reservation and Staff Management interfaces
- Hotel Dashboard

2. Business Logic & Automation Layer

- Salesforce Flow
- Validation Rules
- Apex Trigger
- Batch Apex

3. Data Layer

- Account – Hotel
- Contact – Guest
- Room
- Reservation
- Staff
- Room Service
- Payment

4. Reporting & Analytics Layer

- Rooms by Type
- Occupancy
- Revenue
- Staff Performance
- Total Hotels
- Total Reservations



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- Available Rooms
- Occupancy Trend
- Hotel Reservation Dashboard

The architecture allows hotel data to be stored centrally while automation and Apex handle business rules and reports and dashboards provide operational insights.

Final Project Working Screenshots

The final project screenshots demonstrate the working Salesforce application and its major components. The screenshots can include:

1. **Hotel and Room Management** – Shows hotel details and room information.

The screenshot displays the Salesforce interface for the 'Hotel Reservation Management' application. The top navigation bar includes 'Accounts', 'Rooms', 'Contacts', and 'Reservations'. The main content area shows the details for 'Room 101'. The 'Details' tab is active, displaying a list of fields: Room Number (101), Room Type (Single), Room Price (₹2,000), Status (Available), Hotel (Test Hotel), Created By (Nithin goli, 09/09/2026, 10:28 am), and Last Modified By (Nithin goli, 09/09/2026, 10:28 am). The 'Activity' tab is also visible on the right, showing a list of activities with filters for 'All time', 'All activities', and 'All types'. The 'Upcoming & Overdue' section shows 'No activities to show. Get started by sending an email, scheduling a task, and more.' A 'Show All Activities' button is present at the bottom of the activity section.



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2. Guest Management – Shows guest records maintained through Contacts.

Hotel Reservation Manage...

Accounts

Rooms

Contacts

Reservations

Contact

Mr. nithin hdk

Title

Account Name

Phone (2) ▼
67889877

Email
niuhh@gmail.com

Contact Owner
Nithin.goli

Related

Details

Contact Owner

Nithin.goli

Name

Mr. nithin hdk

Account Name

Title

Department

Birthdate

Reports To

Phone

67889877

Home Phone

Mobile

Other Phone

Fax

Email

niuhh@gmail.com

Assistant

3. Reservation Management – Demonstrates creating and managing room reservations.

Hotel Reservation Manage...

Accounts

Rooms

Contacts

Reservations

Reservation

RES-0002

Related

Details

Reservation Number

RES-0002

Room

101

Guest

nithin hdk

Check In Date

09/09/2026

Check Out Date

09/10/2026

Status

Confirmed

Created By

Nithin.goli, 09/09/2026, 10:31 am

Owner

Nithin.goli

Last Modified By

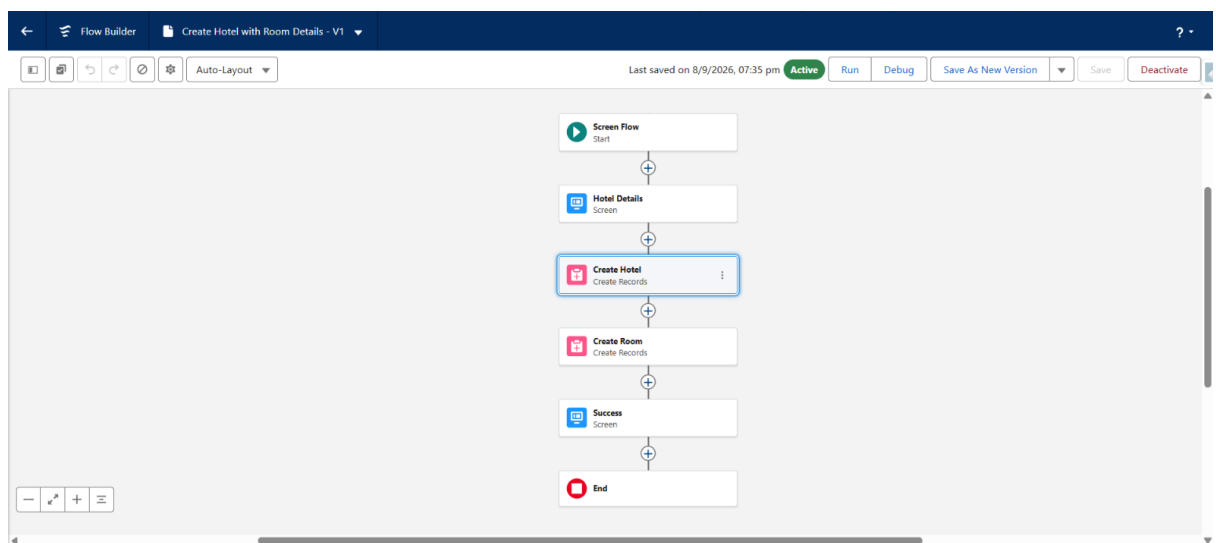
Nithin.goli, 09/09/2026, 10:31 am



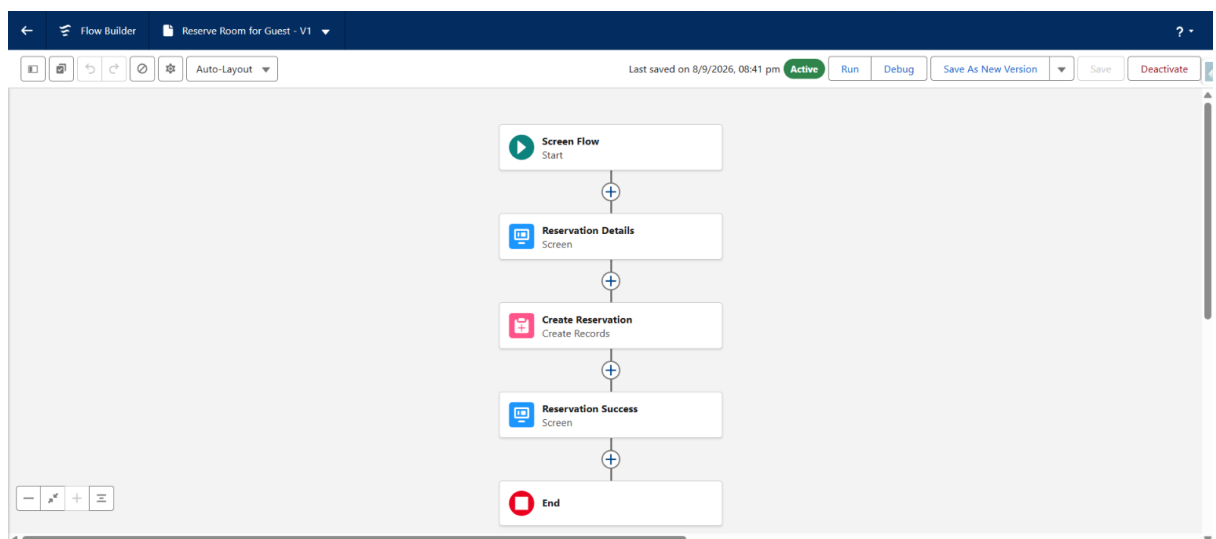
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4.Flow Automation – Shows the configured Salesforce Flows for reservation and payment processes.

--Create Hotel with Room Details flow



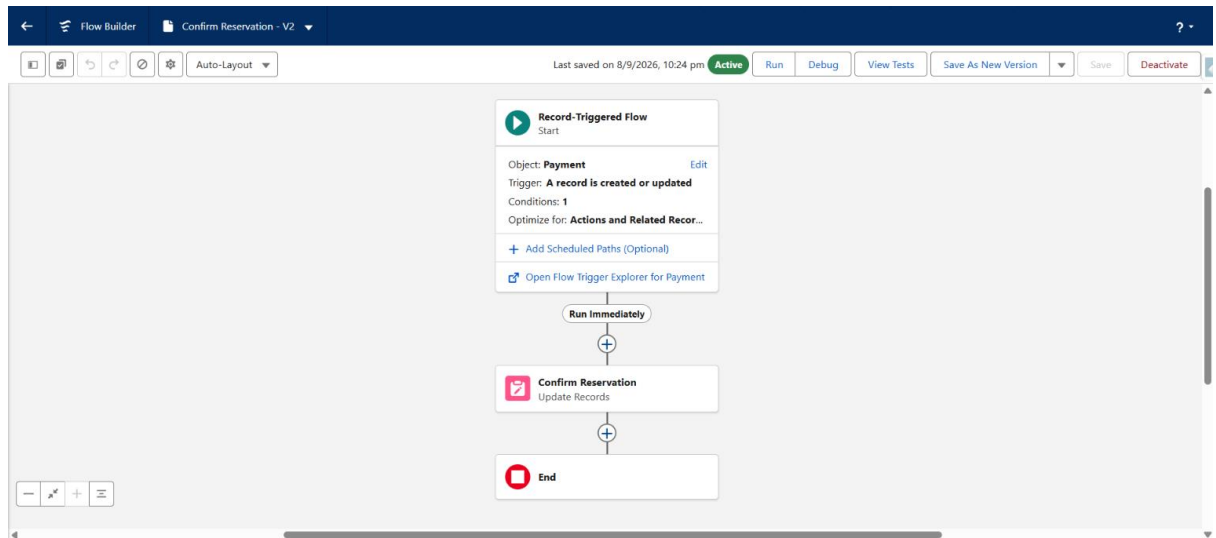
--Reserve Room for Guest flow



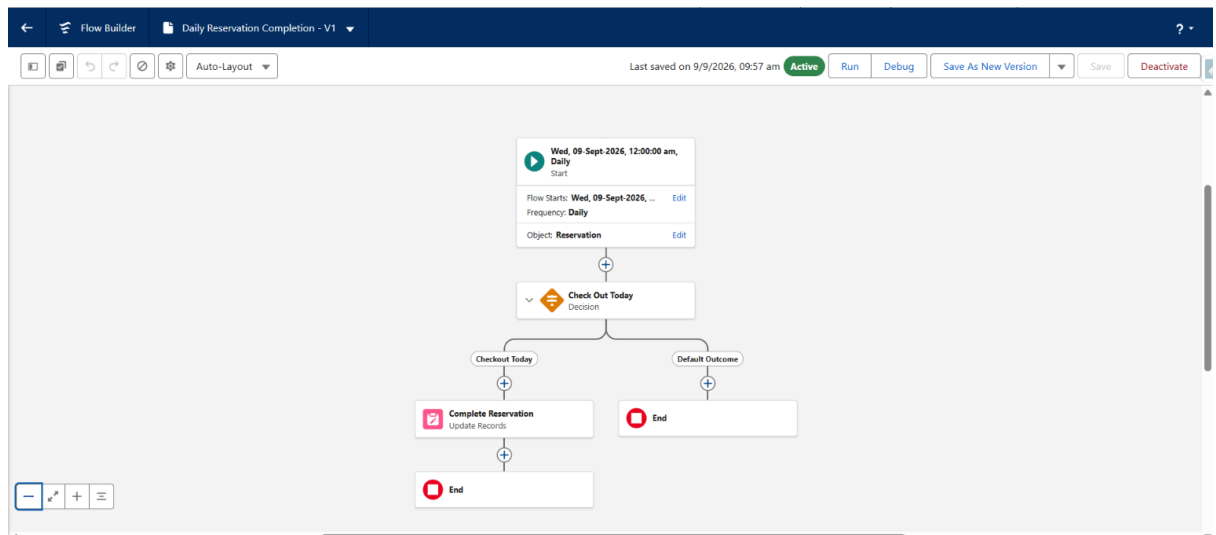


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--Reservation Created → Confirmed flow



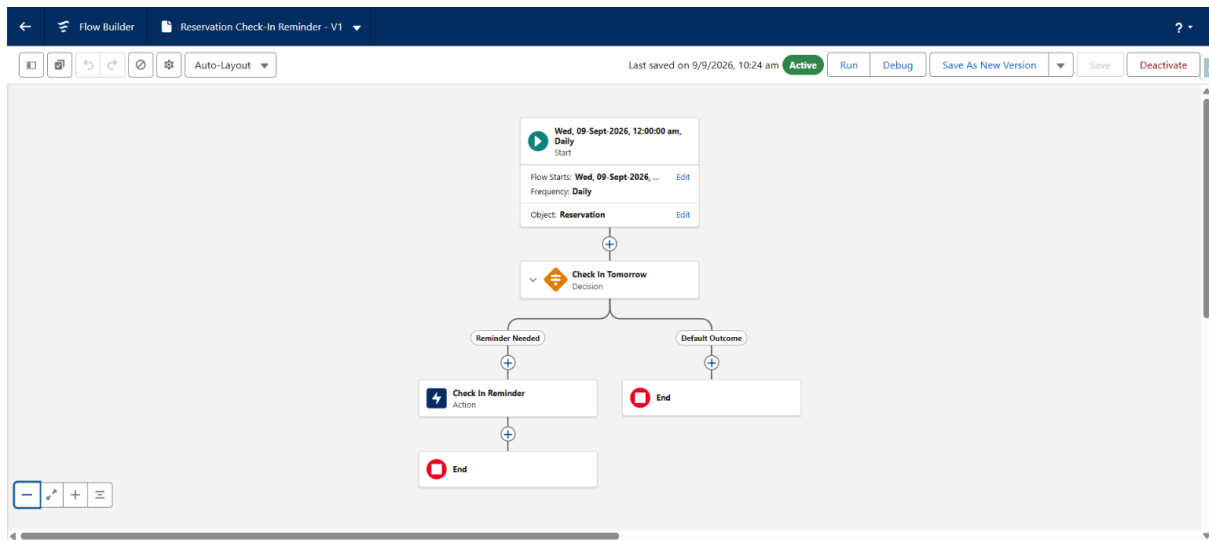
--Daily Reservation Completion





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--Check-In Reminder



5.Lightning Web Components – Shows the Hotel Dashboard, Reservation, and Staff Management components.

Hotel Reservation Management - Reservations

Accounts Rooms Contacts Reservations Payments Room Services Staffs

Reservations

Recently Viewed

1 item • Updated a few seconds ago

Search this list...

	Reservation Number
1	RES-0002

Hotel Reservation Management - Staffs

Accounts Rooms Contacts Reservations Payments Room Services Staffs

Staffs

Recently Viewed

3 items • Updated a few seconds ago

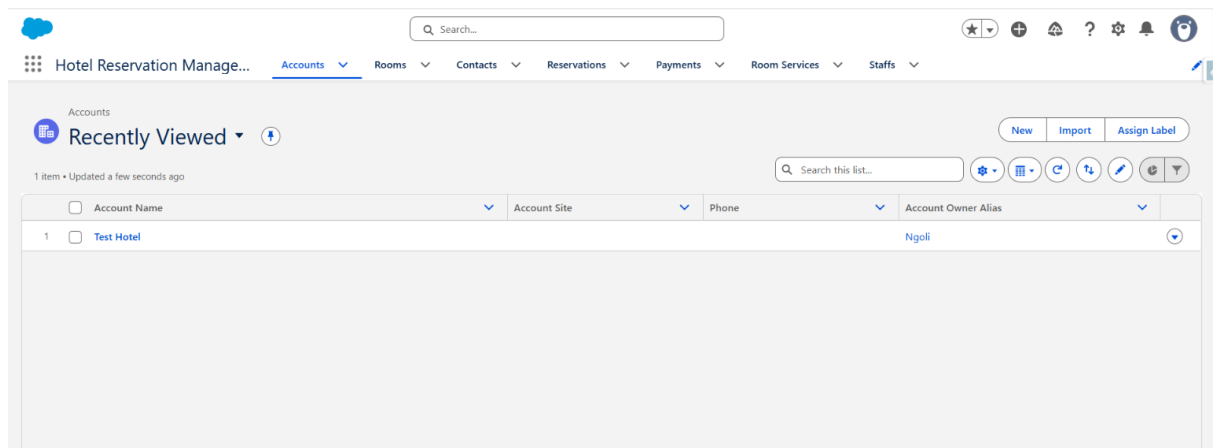
Search this list...

	Staff Name
1	king
2	rani
3	raju



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6,Dashboard – Shows the overall Hotel Reservation Dashboard with important hotel metrics.



The screenshot shows a web application interface for "Hotel Reservation Management". The top navigation bar includes a search bar, a star icon, a plus icon, a refresh icon, a question mark, a settings gear, a bell, and a user profile icon. The main navigation menu lists: Hotel Reservation Manage..., Accounts, Rooms, Contacts, Reservations, Payments, Room Services, and Staffs. The "Accounts" section is active, displaying a "Recently Viewed" list. The list contains one item, "Test Hotel", with a checkbox and a link to "Ngoli". The table has columns for Account Name, Account Site, Phone, and Account Owner Alias.

	Account Name	Account Site	Phone	Account Owner Alias
1	☐ Test Hotel			Ngoli



New Learnings

Through the development of the Hotel Reservation Management System, I gained practical knowledge in Salesforce administration and development. The major learnings include:

- Understanding Salesforce standard and custom objects.
- Creating fields, relationships, and data models.
- Using Account and Contact objects for hotel and guest management.
- Creating Validation Rules and Record Types.
- Designing and implementing Salesforce Flows.
- Understanding Apex programming and Trigger development.
- Implementing duplicate room booking prevention using Apex.
- Learning Batch Apex for processing reservation information.
- Developing Lightning Web Components using Visual Studio Code.
- Creating and customizing Salesforce Reports and Dashboards.
- Understanding role hierarchy, profiles, permissions, and data security.
- Improving debugging and problem-solving skills.

Management and Professional Learnings

The project also helped me improve my:

- Planning and organizing project activities.
- Time management and task prioritization.
- Problem-solving and debugging abilities.
- Technical documentation skills.
- Understanding of real-world business requirements.
- Ability to test and improve application functionality.
- Confidence in learning and implementing new technologies independently.

Overall Experience

The overall experience of developing this project was highly valuable and informative. I gained hands-on experience in building a real-world application using the Salesforce platform. During the project, I faced challenges while configuring objects, relationships, Flows, Apex, and Lightning Web



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Components. Solving these challenges improved my technical knowledge and troubleshooting abilities.

The project helped me understand how Salesforce can be used to automate business processes and manage data efficiently. It also improved my confidence in Salesforce administration, development, reporting, and application design. Overall, this project was an important learning experience that strengthened my technical, analytical, and professional skills.



Conclusion and Future Scope

Conclusion

The Hotel Reservation Management System was developed successfully using the Salesforce platform to manage hotel operations in a centralized and efficient manner. The project achieved its main objectives, including hotel, room, guest, reservation, staff, room service, and payment management.

Salesforce Flows were used to automate reservation confirmation, payment-based status updates, reservation completion, and check-in reminders. Apex was implemented to prevent duplicate room bookings and support occupancy-related processing. Lightning Web Components, Reports, and Dashboards provided an interactive interface and useful insights into hotel operations.

Overall, the project improved data accuracy, reduced manual work, and provided practical experience in Salesforce administration, development, automation, and reporting.

Future Scope

The project can be further enhanced by implementing the following features:

- Integration with online payment gateways.
- Email and SMS notifications for guests.
- Online room booking and cancellation.
- Customer self-service portal.
- Mobile application support.
- Advanced occupancy and revenue analytics.
- Hotel review and rating system.
- Integration with external hotel booking platforms.
- AI-based hotel and room recommendations.
- Multi-hotel and multi-location management.
- Enhanced security and customer data protection.
- Integration with external accounting and billing systems.



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